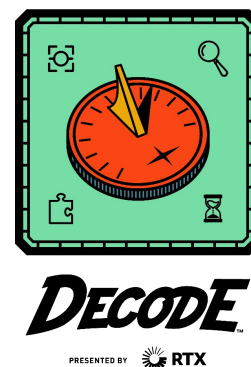
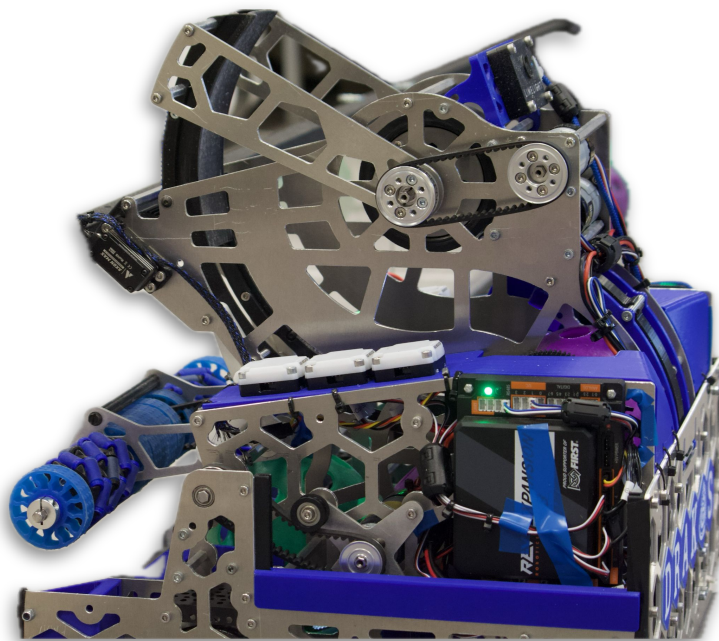


Team #25619 Nova Pyra
Engineering Portfolio | 2025-2026



Team History

Rookie Year: **2024** (Now in **2nd season!**)

The sister team of **FRC S.S. Prometheus!**

Winner of the **Louisiana** Regional Competition **2025** and **2026**
Winning Alliance **Captain** 2026

The name “Nova Pyra,” meaning “New Fire,” reflects the team’s origin story – a revival of robotics at **Mandeville Junior High**, inspired by the myth of Prometheus, who granted fire to humanity.

Mission Statement

Nova Pyra is committed to advancing the boundaries of innovation in robotics, fostering creativity, teamwork, and technical excellence. Together, we strive to ignite a passion for engineering and technology.



Find These... New Members and Student Leadership!

Outreach and Media: Pg. 5, 7-8

Robot & Strategy:
Pg. 9-11

Engineering Process: Pg. 12-14

From our outreach events we get new team members every year! In order to **teach** and include them in all aspects of robotics, we share various aspects of team jobs and allow them to focus on their specific **interests**, whether in build, CAD, code, or outreach.

Returning members are invited to try out for **leadership** positions! Leadership roles are earned through demonstrated initiative, technical contribution, and team impact.



Members

On our team, we value work equity for every team member.



Team Captain, all-rounder, leadership, time management, and scheduling
Personal Goal: Create an environment where anyone feels comfortable to confide in me.



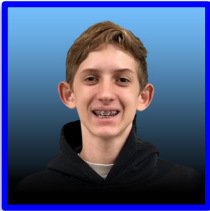
Build Lead, creating a build schedule, overseeing construction
Personal Goal: Develop the technical skills to create ideas that are nearly impossible to create.



Documentation Lead, creating judging materials (Engineering Portfolio and presentation)
Personal Goal: Learn more about the engineering process.



Media Lead, curating our presence in our community and online
Personal Goal: Get better at communicating with people I'm not familiar with.



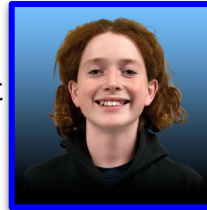
Code Lead, writing code for autonomous and tele op
Personal Goal: Become a better problem solver.



CAD Lead, utilizes OnShape to design robot parts
Personal Goal: Be able to talk better in front of people and perform better under pressure.



Pit Lead/Build Member, organize and staff pit area; assist with robot construction and testing
Personal Goal: Become better at communicating what I want/need.



CAD/Build Member, assists with robot design and construction
Personal Goal: Make friends, grow my knowledge of engineering and have fun.



Build Member, focuses on robot construction
Personal Goal: Get better at computer design.



Media/Documentation member, assists with outreach, fundraising, and documentation.
Personal Goal: Make more friends!



Website/Media Member, updates and maintains team website
Personal Goal: Be able to better interact with people and communicate my ideas.



Code/Media Member, assists with code development and fundraising.
Personal Goal: Learn more about engineering.



Skill Development

Developing student **skills** with **student leaders** and mentors.



- We host sessions with our **student leaders** as well as our **recruited mentors** in order to teach new students the skill areas they want to pursue on our team.
- **Hands-on learning** is valuable during our short season.

Build Team

Goal: Manage the design and physical **construction** of the robot.

- Age can cause risk when using and learning tools → Mentors have to supervise all usage of power tools to ensure safety.

Captain

Goal: **Communicate** and organize team plans from members to leaders.

- Communication from mentors to leads can be difficult or unclear → Captain works with coach to ensure there are multiple ways of communicating.

CAD Team

Goal: Manage the 3D **design** of the robot using CAD software (Onshape).

- Inexperience with CAD programs can slow down the idea to prototype process during the season → Pre-season training allows for more experience when it counts.

Documentation Team

Goal: **Maintain** the Engineering Portfolio and judging materials.

- Financial information and performance data is shared through booster members → Easy communication of data and financial plans from the team leads to booster members.

Code Team

Goal: Manage robot code **development**, learning Java and FTC SDK standards.

- Inexperience with coding programs can slow down ideas to take shape during the season → Pre-season training allows for faster results later.

Media Team

Goal: Manage the team's **outreach** and marketing design.

- Age restrictions in the parish bar younger members from reaching out through social media → all outreach is closely supervised through mentors.

To **teach** new members and gain team **efficiency** through long-term **planning**.

- Mentors and team members select students who excel in certain areas, demonstrate strong **leadership skills**, and who can work well with others
- Our student leaders learn **communication** skills and **gracious professionalism**

Developing Student Leadership



Recruiting mentors, coaches, volunteers, and new teams is essential for our team to grow the **FIRST Pipeline** by attending events and reaching out to members of the STEM community or anyone with an interest in robotics.

Recruitment Events

LATM/LSTA Teacher Convention

Event Goal

Reach out to our community to **recruit** mentors, coaches, and volunteers throughout Louisiana



Our **Three** STEM Nights at MES and PES taught students and teachers about the **FIRST community** and their involvement in robotics, in an effort to create **new teams** and recruit new **students** and **mentors**.

LATM/LSTA Teacher Convention strengthened the **FIRST Pipeline** by discussing **mentoring** opportunities with Louisiana teachers who specialized in STEM subjects.



STEM Nights at Local Elementary Schools

Event Goal

Outreach with local schools to **recruit** new teams and robotics students, coaches, and mentors



Brad Villemeur (mentor):

Guides and teaches students how to use **CAD** software and the fundamentals of **engineering** and **design**. Recruited through sponsorship and connection with DDG.



Ryan Harvey (mentor):

Guides and teaches students how to use **code** software, develop hardware **solutions**, and manage **finances**. Recruited through partnership with **FRC 2992**.



Laura Meche (mentor):

As a former judge, she guides our **documentation** team as well as providing guidelines of the **engineering process**. Recruited through **FIRST** collaboration event.

Successful Recruitment of Mentors, Teams, Coaches, Volunteers, etc...



Amy Bordelon (Coach): Head of media, documentation, and finances as one of 2 coaches on our team. Recruited through **STEM community**.



Matthew Gilbert (mentor): Teaches student about the **engineering process** and the **construction** of the robot. Recruited through STEM community.



Financial Stability Plan

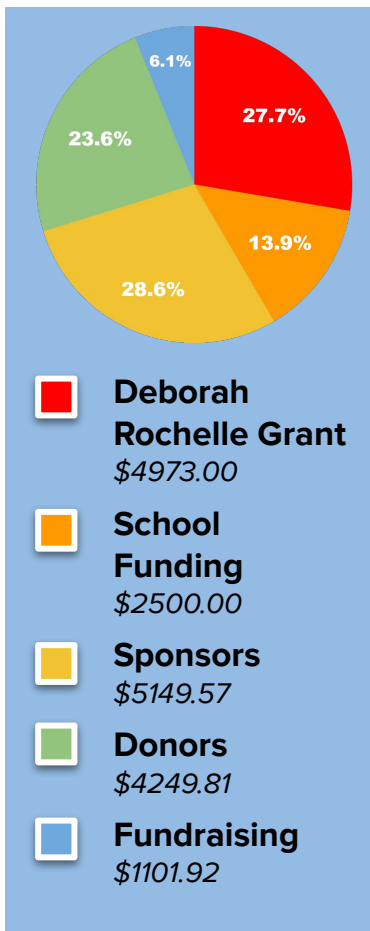
Financial planning is crucial for **sustaining** team spending and maintaining a balanced budget.

Our **financial sustainability plan** includes

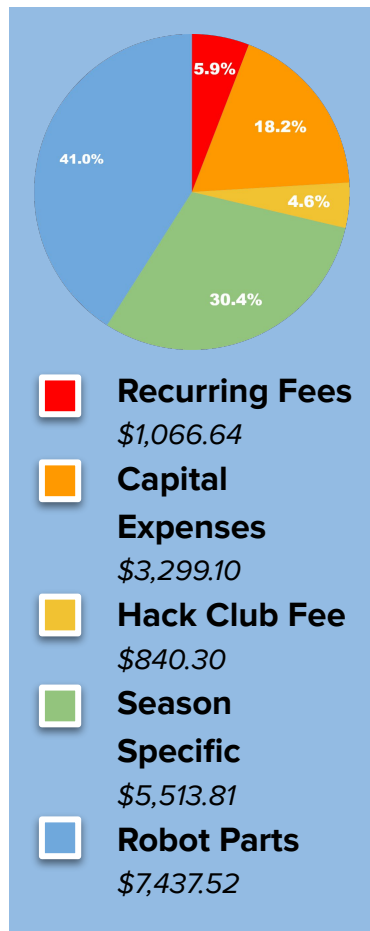
- **Diversifying income sources:** grant writing, local business sponsorships, seasonal fundraisers, and long-term agreements with repeating sponsors
- **Tracking income and expenses** using a Google Sheet updated weekly
- **Ensuring financial stability in the future** by setting aside funds for next season

Spending Statistics

Income Breakdown



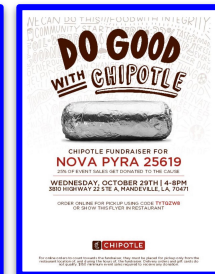
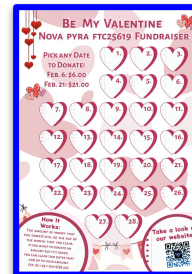
Expenses Breakdown



*Balance from previous season of \$1050

Fundraisers

This year we hosted many fundraising events, not only to raise money for our team, but to also **reach out** to our community and share our passion for robotics. These fundraisers were held with **local businesses** and on our school campus.



Learning to **manage** finances as a team is a critical step in teaching students the fundamentals of **FIRST**.



Collaborations with the Engineering Community

Visiting DDG and Intralox

Event Goal

Learn more about the **engineering process** and explore engineering and **STEM careers**.

Learned about engineering **job opportunities** and the **processes** real engineers use every day to develop **solutions**.



Discussed **mentoring** opportunities with **FLL** teams. Shared information about **FIRST** to adults & students interested in involvement. Developed **connections** with schools and companies in STEM.

NOLA STEM Fest

Event Goal

Outreach to our engineering community and develop new **STEM connections**.

Ongoing Communication with DDG: Besides our visit to DDG, our team is in constant communication with DDG through our **mentor** Brad Villemeur. We learn valuable knowledge of CAD and engineering as a whole. We receive support, resources, and mentorship from them, making them one of our most important sponsors and **active connections**.

Mentors in STEM

Laura Meche: an expert in **technical writing** at **Turner Industries**, she offers advice on our portfolio and presentations

Matthew Gilbert: a **research scientist** with the **USDA**, he works hands-on with students to teach building skills

Ryan Harvey: a **professional coder** and college professor, he helps students become proficient in coding as well as serving as Booster Club president.

These Connections Teach Us...

- Learning about possible **STEM careers** our team members may pursue in the future
- Gaining valuable insight on the **engineering process** that real engineers and scientists use every day to further develop our own prototyping and game strategy
- Building confidence and courage to talk to others and communicate as **gracious professionals** at competitions and other events
- Incorporating the **ideals** of STEM into school and robotics



Collaborations with the FIRST Community



Collaborated with **sister FRC team** S.S. Prometheus at a hosted event where our team recruited and spread information for FTC and the upcoming season.

Northshore Knock Out

Event Goal

Reach out to the **FIRST** community (**FRC** specifically) and share information about **FTC**.

FTC Scrimmage

Event Goal

Improve **driving** practice while participating in matches with many teams to improve **strategy**.

Our drive team participated in mock-game matches before **state** with 3 other FTC teams and collaborated on game strategy and our **engineering process** this season.



NOLA STEM Fest (see details on Pg. 7) we attended with **FIRST** and other **FTC** teams.

FIRST Partnerships

FTC 6448 (Jesuit Blue Jays)



We collaborated in **driver practice** before our state competition and discussed engineering and **judging** strategies.

FRC 2992 (S.S. Prometheus)



Our **sister** team shared a shop with us, teaching us how to use complex **machinery** and navigating CAD & code.

- Our social media allows us to connect with people around the world in order to increase the **FIRST** pipeline
- It also allows us to **collaborate** with other teams and post our **strategies** while learning new ones from others

Social Media!



Creative Outreach

Our team continues to make efforts in creative outreach. We brainstorm and **research** different ways to actively reach out, even to people **not** in the STEM community, but to students and adults everywhere.



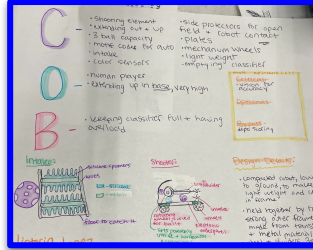
COB

Each year we start with a COB strategy session to create a game strategy.

Critical → Classify Artifacts & Park

Optional → **Double Park**

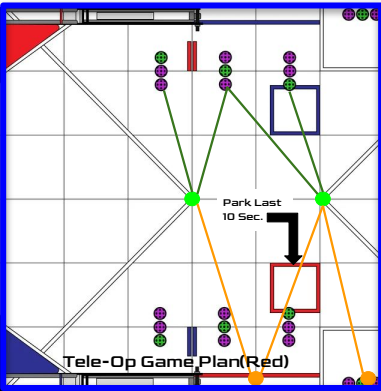
Bypass → Score Depot & Pattern



Design Choices

- High-speed shooter & intake system
- Fast, efficient drivetrain for rapid movement
- Ability to score from anywhere on field

Game Strategy Engineering Design Process



AUTO: Developed 7 autonomous routines (14 total across both alliance colors), including near and far variants. Paths are adjusted based on alliance partners and starting configurations.

TELEOP: Capable of **scoring from any position on the field**. Positioning (near/far) is dynamically adjusted based on alliance partners, with a **focus on fast cycle times** and maximizing artifacts scored in the classifier.

ENDGAME: Prioritize a full park, scoring additional artifacts if time allows, while bypassing the pattern when not beneficial.

Having a detailed process allows us to **sustain** effective prototyping and testing periods.

Ask : Identify the need and **constraints**

- Used our COB strategy at the start of the season to **define priorities** and guide game strategy decisions.

Imagine : Investigate the problem

- Analyzed **risks, constraints, and gameplay challenges** to inform design decisions.

Make a Plan : Select a promising solution

- **Developed concepts** using CAD and hand-drawn designs to evaluate potential solutions.

Create : Build a prototype

- Built **iterative prototypes**, beginning with cardboard, progressing to 3D-printed parts, and refining into final designs.

Test : Evaluate the prototype

- **Tested prototypes in game-like conditions** to measure consistency, efficiency, and performance.

Improve : Redesign as needed

- **Refined designs through repeated iterations**, using testing data to drive continuous improvement.



Using the Engineering Process

Iterations

The Engineering Process is crucial for **developing solutions** and **improving** the intake.

Ask : Identify the need and constraints

- Needed a **fast, reliable intake** system capable of smooth artifact collection while **preventing jamming**.

Imagine : Investigate the problem

- Investigated **intake geometry**, roller placement, and **artifact behavior** to **identify causes of jamming** and inconsistent pickup.

Make a Plan : Select a promising solution

- Used **CAD** (Onshape) to design and iterate intake configurations, focusing on **roller placement, artifact control, and feed direction**.

Create : Build a prototype

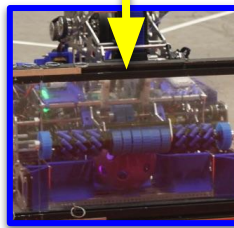
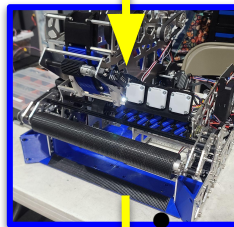
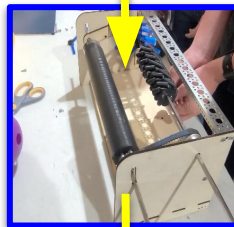
- Began prototyping with **cardboard designs** for rapid iteration and concept validation.
- Transitioned to **wood structures** to test motor-driven systems with improved durability.
- Finalized the design using **metal components and custom-molded silicone wheels** for strength, grip, and consistent performance

Test : Evaluate the prototype

- Conducted **repeated testing** cycles, recording data on **intake speed, consistency, and jamming frequency** to guide design improvements.

Improve : Redesign as needed

- To **improve grip and artifact control**, **custom intake wheels were designed and manufactured** using CAD and 3D-printed molds.
- Additional **vector wheels** were added to **center artifacts** and ensure **consistent feeding into shooter**.



- Began with **cardboard prototypes**, testing multiple intake designs including different wheel types and roller configurations to evaluate artifact capture and control.

- Transitioned to a **wood prototype**, selecting a **single front roller with grip tape** for initial intake, paired with a **rear roller using vector wheels** to direct artifacts into the shooter feed system.

- Through testing, the grip tape roller was replaced with **custom silicone wheels**, significantly improving grip and consistency.

- Further iterations led to simplifying the system by **merging the two-roller design into a single roller**, reducing complexity and improving reliability.

- The final intake uses a **combined roller system**, with **vector wheels on the outer edges to center the artifacts** and **custom silicone wheels in the middle** to drive them directly into the shooter feed.

Custom Intake Wheels

Our **custom-molded intake wheels** improve pickup speed and consistency on the field. They were manufactured using **3D-printed molds and cast silicone**, allowing us to optimize grip and performance.



Using the Engineering Process

Iterations

The Engineering Process is crucial for **developing solutions** and **improving** the shooter.

Ask : Identify the need and constraints

- Needed a **fast, consistent shooter** capable of accurate, repeatable scoring while remaining simple and driver-friendly.

Imagine : Investigate the problem

- Used **rapid prototyping** to visualize ideas, identify issues, and explore different shooter configurations early in development.

Make a Plan : Select a promising solution

- Designed iterations in **CAD** and tested multiple configurations through **3D printed prototypes**, focusing on optimizing **wheel compression and contact** with the artifact.

Create : Build a prototype

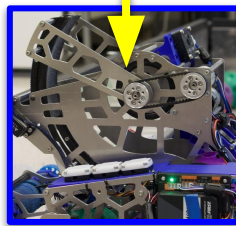
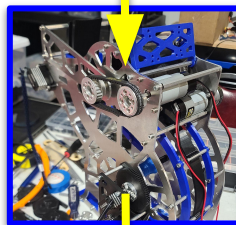
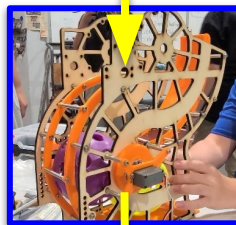
- Initial prototypes were 3D printed, allowing **flexibility in testing** angles, shot behavior, and speed.
- Later iterations transitioned to wood structures to **improve stability and consistency** in artifact transfer to shooter.

Test : Evaluate the prototype

- **Tested** the shooter in real driving conditions using a modular chassis, **recording data** on shot timing, accuracy, and consistency to guide improvements.

Improve : Redesign as needed

- Refined the system to enable fast, repeatable shooting, with the programming team developing a lookup table system to **dynamically set flywheel RPM and hood angle** based on distance to the target, while **odometry provides precise, repeatable robot alignment** for accurate aiming.



- Began with **3D printed prototypes** to quickly test motor speed, flywheel behavior, and compression.
- Tested **fixed-angle flywheel** configuration to **evaluate and establish baseline RPM performance** and shot consistency.
- Transitioned to a wood structure to improve vertical alignment and system stability during repeated testing.
- Developed an **S-shaped intake to shooter path** to improve artifact flow and ensure smooth, consistent feeding.
- Refined the system into a **metal "serpent" structure**, guiding artifacts from the intake using controlled contact from a **custom rubber wheel** into the shooter.
- Finalized the design with an **adjustable hood**, allowing consistent shots from multiple angles and field positions.

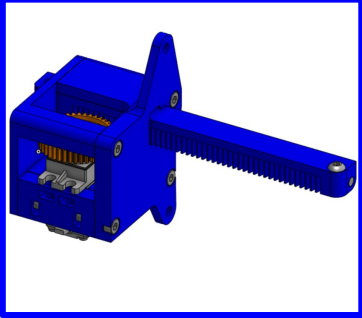
Custom Hood

Our adjustable custom hood enables **consistent, repeatable shooting** from any position on the field, using **Limelight pose updates** and **odometry-based targeting** to accurately aim at the goal.



Constraints

Combating risks is a big part of the **engineering design process**.



Risk:

Intake was able to carry **four balls into the intake** without driver awareness, creating risk of penalties due to holding excess artifacts.

Our Solution:

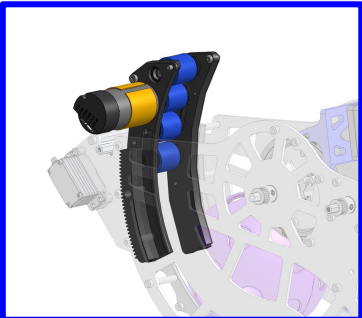
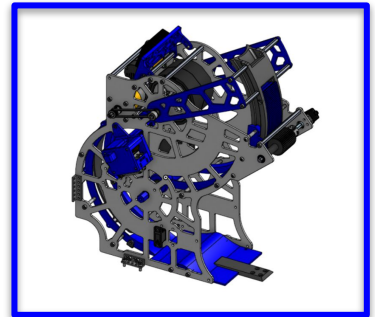
A **rack and pinion feed stop** acts as a **hard stop** for the first incoming artifact, defining a fixed position within the intake. This limits available space for additional artifacts, **preventing a fourth artifact from being collected**.

Risk:

At higher flywheel speeds, **artifacts gained excessive backspin along the hood interface**, resulting in bounce-outs, reduced consistency, and **less than 75% shot accuracy**.

Our Solution:

We tested multiple hood and flywheel interactions, **iterating on compression and material contact** to reduce backspin, resulting in improved shot consistency.



Risk:

In order to reduce backspin, we tried to add **kicker wheels**, but experienced inconsistencies in tuning and shot performance.

Our Solution:

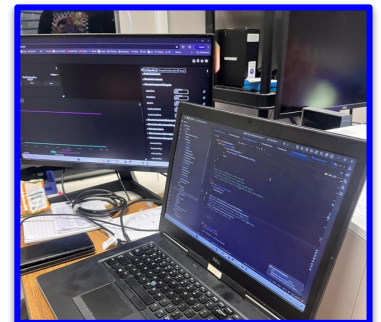
Learning that sometimes, **newer isn't better**. After testing, we determined that added complexity reduced reliability. We removed the kicker wheels and refined our original design for more consistent performance.

Risk:

Without proper tuning, **shot speed varied** between cycles, leading to **inconsistent scoring** and reduced reliability.

Our Solution:

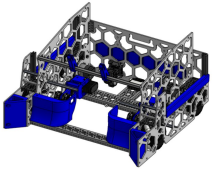
We implemented **RPM-based PID control** to maintain consistent flywheel speed. This stabilized shot performance, reduced variability, and allowed for quick adjustments between shots.



Our **final design** is achieved by continuously **testing, evaluating, and refining solutions** to each identified risk.

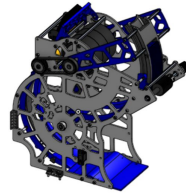


Finalized Designs



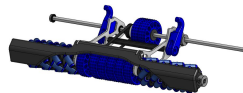
Drivetrain

- Enables precise positioning and **smooth navigation** during both auto and tele-op
- Metal plated, **protected sides**
- Easy **access to wiring**



Shooter

- Designed to **maximize energy transfer**
- 3D printed **custom hood**
- Custom **steel inserts** to **increase flywheel inertia**



Intake

- Designed for **fast, controlled intake** into shooter system.
- Custom **3D printed vector wheels**
- Custom **molded silicon wheels**
- Angled compression path
- Mechanical **feed stop**

Reaching Efficiency

Using our **engineering process**, we **tested** each system of our robot to reach maximum efficiency, and **improved** when needed.

- Through our **limelight** camera and **PID** code, our shooter achieves high accuracy.
- A gate transfer system allows artifacts to **feed quickly and consistently**, preventing jams and maintaining continuous scoring cycles.
- Testing data was used to refine **flywheel RPM**, ensuring consistent shot performance.
- **Sensor feedback** and **control logic** ensure shots are only taken when conditions are optimal, reducing wasted cycles and increasing scoring efficiency.

[illegible]

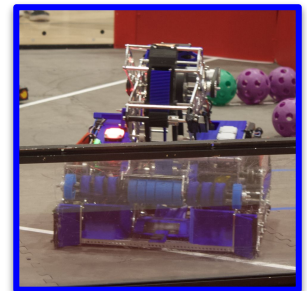
RPM	D. Stone
2470	342
2430	326
2550	375
2640	385

2640	385
2550	375
2470	342
2430	326

Consistent with Game Strategy

Auto:

Coordinate with alliance partners to select from **seven autonomous paths**, maximizing scoring.



Tele-Op:

Shoot from any launch line,
allowing flexible movement
and strategic positioning
around defense.



End Game:

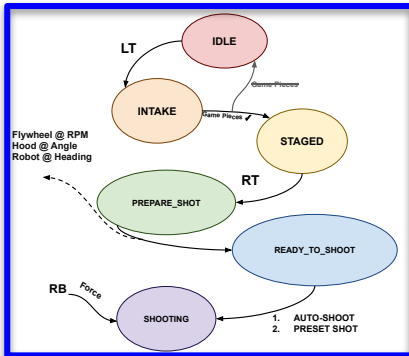
Score final shots and secure parking, **prioritizing reliability** over pattern completion.

With a flexible game strategy, we make sure our robot is **suitable** for all alliance selections to **maximize** capabilities.



Shot Controller State Machine

A **custom** Shot Controller state machine to **automate** our entire shooting cycle. The state machine **reduces driver workload** by automating timing, alignment, and shooting decisions.

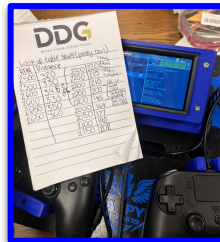


Our robot operates using a **state-based control system**, transitioning between defined states based on **sensor feedback and driver input**. It uses Limelight data to refine robot pose and **odometry to calculate distance and alignment to target**.

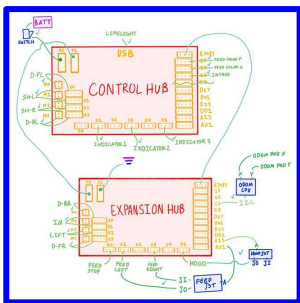
Testing

Data-Driven Refinement

Code tuning is performed through in **repetitive, data-driven testing on the field**. For lookup tables, the robot is placed in a fixed position and the same shot is repeated over 50+ times to evaluate **accuracy, timing, and positioning**.



This data is then used to refine system performance through:



- External feedback loops
- Sensors integration that improve consistency
- Control algorithms for coordinated system behavior
- Color sensors to monitor artifact flow
- A feed control algorithm that prevents double-feeding and only fires when flywheel RPM stabilizes.
- PID control for maintaining consistent flywheel RPM control.

Using the Engineering Process

- Ask :** Identify the need and constraints
- **Need:** Create a **driver-friendly control** system that reduces **button complexity** while maintaining **fast, consistent shooting**.
- Imagine :** Investigate the problem
- Brainstorm **automated control strategies**, including **subsystem coordination** and **sensor integration**, to support early iterations and testing.
- Make a Plan :** Select a promising solution
- Developed a **mecanum drivetrain code base** as a foundation and designed a **controller state machine** architecture to simplify **button mapping** and **automate actions**.
- Create :** Build a prototype
- Implemented an **initial state machine** using simple **integer-based logic** to map system behavior, allowing for **structured tuning** and expansion later in the season.
- Test :** Evaluate the prototype
- Collected **performance data** through **repeated testing** and lookup tables, which were later integrated with PID tuning to improve consistency and accuracy.
- Improve :** Redesign as needed
- Refined **control logic** and updated lookup tables through **iterative testing**, improving **shot accuracy** and overall **system reliability** throughout the season.